

# Yuntian DENG

Google Scholar◇ Personal Website

yuntian@uwaterloo.ca

## Employment

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|----------------------------------------------------------|-------------------|
| University of Waterloo                                   | Waterloo, Canada  |
| Assistant Professor, Cheriton School of Computer Science | Aug 2024–Present  |
| Vector Institute                                         | Toronto, Canada   |
| Faculty Affiliate                                        | Aug 2024–Present  |
| Harvard University                                       | Remote            |
| Associate, Computer Science                              | Jun 2024–Present  |
| NVIDIA                                                   | Toronto, Canada   |
| Visiting Professor, Yejin Choi’s team                    | Nov 2024–Nov 2025 |
| Allen Institute for Artificial Intelligence              | Seattle, WA       |
| Postdoc (advised by Yejin Choi)                          | Jul 2023–Jul 2024 |

## Education

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|---------------------------------------------------|----------------|
| Harvard University                                | Cambridge, MA  |
| Ph.D. in Computer Science                         | May 2023       |
| Advisors: Alexander Rush and Stuart Shieber       |                |
| Carnegie Mellon University                        | Pittsburgh, PA |
| Master in Language Technologies                   | Aug 2016       |
| Advisor: Eric Xing                                |                |
| Tsinghua University                               | Beijing, China |
| Bachelor of Engineering, Department of Automation | Jul 2014       |

## Research Interests

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Program-as-Weights and neural programs; neural world models and interactive systems; language-model reasoning and implicit computation; real-world LLM usage at scale; open research tools and datasets.

## Awards

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|                                       |          |
|---------------------------------------|----------|
| Laude Institute Slingshots (NeuralOS) | Jun 2026 |
| Google Research Award (co-PI)         | Apr 2026 |
| Argonne National Lab Impact Award     | Jan 2023 |

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|-----------------------------------------------------------------------------|-----------------------------------|
| University of Chicago Rising Stars in Data Science                          | Nov 2022                          |
| ACM Gordon Bell Special Prize for COVID-19 Research                         | Nov 2022                          |
| NVIDIA Fellowship                                                           | Dec 2021                          |
| Microsoft Turing Academic Program, Selected for Improving LM Reasoning      | Dec 2021                          |
| Harvard Certificates of Distinction in Teaching                             | Spring 2019, Fall 2020, Fall 2021 |
| Twitch Fellowship Finalist                                                  | Jan 2021                          |
| DAC 2020 Best Paper Award                                                   | Jul 2020                          |
| Baidu Fellowship                                                            | 2019                              |
| French American Doctoral Exchange Program (French Embassy, 10 US laureates) | Jun 2018                          |
| ACL 2017 Best Demo Paper Award Runner-Up                                    | Aug 2017                          |

## Publications and Preprints

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\*Equal contribution. <sup>†</sup>Equal advising.

### Selected Papers

- 48 Wentao Zhang\*, Xuanhe Pan\*, Han Zhou\*, Yang Lu<sup>†</sup>, Yuntian Deng<sup>†</sup>. Interactive Training 2: Auditable Control Plane for Live Model Training. arXiv 2026
- 47 Wentao Zhang\*, Liliana Hotsko\*, Woojeong Kim\*, Pengyu Nie, Stuart Shieber, Yuntian Deng. Program-as-Weights: A Programming Paradigm for Fuzzy Functions. arXiv 2026
- 46 Luke Rivard, Sun Sun, Hongyu Guo, Wenhui Chen, Yuntian Deng. NeuralOS: Towards Simulating Operating Systems via Neural Generative Models. ICLR 2026
- 45 Wentao Zhang, Yang Young Lu, Yuntian Deng. Interactive Training: Feedback-Driven Neural Network Optimization. EMNLP 2025 Demo
- 44 Wentao Zhang, Woojeong Kim, Yuntian Deng. From Chat Logs to Collective Insights: Aggregative Question Answering. EMNLP 2025 Main Oral
- 43 Yuntian Deng, Yejin Choi, Stuart Shieber. From Explicit CoT to Implicit CoT: Learning to Internalize CoT Step by Step. arXiv 2024
- 42 Yuntian Deng, Wenting Zhao, Jack Hessel, Xiang Ren, Claire Cardie, Yejin Choi. WildVis: Open Source Visualizer for Million-Scale Chat Logs in the Wild. EMNLP 2024 Demo
- 41 Bill Yuchen Lin, Yuntian Deng, Khyathi Chandu, Faeze Brahman, Abhilasha Ravichander, Valentina Pyatkin, Nouha Dziri, Ronan Le Bras, Yejin Choi. WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild. ICLR 2025 Spotlight

- 40 Wenting Zhao, Xiang Ren, Jack Hessel, Claire Cardie, Yejin Choi, Yuntian Deng. WildChat: 1M ChatGPT Interaction Logs In The Wild. ICLR 2024 Spotlight; used to evaluate OpenAI’s o1 and Anthropic’s Claude 3; covered by Washington Post and NZZ
- 39 Yuntian Deng, Kiran Prasad, Roland Fernandez, Paul Smolensky, Vishrav Chaudhary, Stuart Shieber. Implicit Chain of Thought Reasoning via Knowledge Distillation. arXiv 2023
- 38 John X. Morris, Chandan Singh, Alexander Rush, Jianfeng Gao, Yuntian Deng. Tree Prompting: Efficient Task Adaptation without Fine-Tuning. EMNLP 2023
- 37 Yuntian Deng, Volodymyr Kuleshov, Alexander Rush. Model Criticism for Long-Form Text Generation. EMNLP 2022
- 36 Yuntian Deng, Anton Bakhtin, Myle Ott, Arthur Szlam, Marc’Aurelio Ranzato. Residual Energy-Based Models for Text Generation. ICLR 2020
- 35 Yuntian Deng, Alexander Rush. Cascaded Text Generation with Markov Transformers. NeurIPS 2020
- 34 Sebastian Gehrmann, Yuntian Deng, Alexander Rush. Bottom-Up Abstractive Summarization. EMNLP 2018
- 33 Zachary Ziegler\*, Yuntian Deng\*, Alexander Rush. Neural Linguistic Steganography. EMNLP 2019 Oral (Demo: [steganography.live](http://steganography.live))
- 32 Yuntian Deng, Noriyuki Kojima, Alexander Rush. Markup-to-Image Diffusion Models with Scheduled Sampling. ICLR 2023 (Demo: [huggingface.co/spaces/yuntian-deng/latex2im](https://huggingface.co/spaces/yuntian-deng/latex2im))
- 31 Yuntian Deng\*, Yoon Kim\*, Justin Chiu, Demi Guo, Alexander Rush. Latent Alignment and Variational Attention. NeurIPS 2018
- 30 Yuntian Deng, Anssi Kanervisto, Jeffrey Ling, Alexander Rush. Image-to-Markup Generation with Coarse-to-Fine Attention. ICML 2017 (Demo: [im2markup.yuntiaideng.com](http://im2markup.yuntiaideng.com))

#### Journal Papers

- 29 Anton Bakhtin\*, Yuntian Deng\*, Sam Gross, Myle Ott, Marc’Aurelio Ranzato, Arthur Szlam. Residual Energy-Based Models for Text. JMLR 2021

#### Other Papers and Preprints

- 28 Liliana Hotsko, Yinxi Li, Yuntian Deng, Pengyu Nie. Code2LoRA: Hypernetwork-Generated Adapters for Code Language Models under Software Evolution. arXiv 2026
- 27 Sajad Ebrahimi, Nima Jamali, Bardia Shirsalimian, Kelly McConvey, Wentao Zhang, Jalehsadat Mahdavi-moghaddam, Maksym Taranukhin, Maura R. Grossman, Vered Shwartz, Yuntian Deng, Ebrahim Bagheri. DetectZoo: A Unified Toolkit for AI-Generated Content Detection Across Text, Audio, and Image Modalities. arXiv 2026
- 26 Kelly McConvey, Jalehsadat Mahdavi-moghaddam, Nima Jamali, Maksym Taranukhin, Sajad Ebrahimi, Wentao Zhang, Karen Eltis, Yuntian Deng, Maura R. Grossman, Vered Shwartz, Ebrahim Bagheri. The

CIFAR Synthetic Evidence Corpus for Detecting AI-Generated Evidence. arXiv 2026

- 25 Xiaoyan Bai, Itamar Pres, Yuntian Deng, Chenhao Tan, Stuart Shieber, Fernanda Viégas, Martin Wattenberg, Andrew Lee. Why Can't Transformers Learn Multiplication? Reverse-Engineering Reveals Long-Range Dependency Pitfalls. arXiv 2025
- 24 Shayne Longpre, Anka Reuel, Dayeon Ki, Zhiping Zhang, Cathy Mengying Fang, Chuanyang Jin, Jennifer Mickel, Cedric Deslandes Whitney, Niloofar Mireshghallah, Victor Ojewale, Megan Richards, Ariel N. Lee, Alex Pentland, Tianshi Li, Yuntian Deng, Sara Hooker, Mykel Kochenderfer, Sanmi Koyejo. The AI Observatory: A Public Measure of Real-World AI Use. In Submission
- 23 David Acuna, Chao-Han Huck Yang, Yuntian Deng, Jaehun Jung, Ximing Lu, Prithviraj Ammanabrolu, Hyunwoo Kim, Yuan-Hong Liao, Yejin Choi. Long Grounded Thoughts: Synthesizing Visual Problems and Reasoning Chains at Scale. ICML 2026
- 22 Yinxi Li, Yuntian Deng, Pengyu Nie. TokDrift: When LLM Speaks in Subwords but Code Speaks in Grammar. ACL 2026
- 21 Wei Liu, Ruochen Zhou, Yiyun Deng, Yuzhen Huang, Junteng Liu, Yuntian Deng, Yizhe Zhang, Junxian He. Learn to Reason Efficiently with Adaptive Length-Based Reward Shaping. ICLR 2026
- 20 Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, Sara Hooker. The Leaderboard Illusion. NeurIPS D&B 2025
- 19 Yifan Zong, Yuntian Deng, Pengyu Nie. Mix-of-Language-Experts Architecture for Multilingual Programming. LLM4Code 2025
- 18 Wenting Zhao, Tanya Goyal, Yu Ying Chiu, Liwei Jiang, Benjamin Newman, Abhilasha Ravichander, Khyathi Chandu, Ronan Le Bras, Claire Cardie, Yuntian Deng, Yejin Choi. WildHallucinations: Evaluating Long-form Factuality in LLMs with Real-World Entity Queries. arXiv 2024
- 17 Zhangchen Xu, Fengqing Jiang, Luyao Niu, Yuntian Deng, Radha Poovendran, Yejin Choi, Bill Yuchen Lin. Magpie: Alignment Data Synthesis from Scratch by Prompting Aligned LLMs with Nothing. ICLR 2025
- 16 Jinjie Ni, Fuzhao Xue, Xiang Yue, Yuntian Deng, Mahir Shah, Kabir Jain, Graham Neubig, Yang You. MixEval: Deriving Wisdom of the Crowd from LLM Benchmark Mixtures. NeurIPS 2024
- 15 Richa Rastogi, Yair Schiff, Alon Hachohen, Zhaozhi Li, Ian Lee, Yuntian Deng, Mert R. Sabuncu, Volodymyr Kuleshov. Semi-Parametric Inducing Point Networks and Neural Processes. ICLR 2023
- 14 Maxim Zvyagin\*, Alexander Brace\*, Kyle Hippe\*, Yuntian Deng\*, Bin Zhang, Cindy Orozco Bohorquez, Austin Clyde, Bharat Kale, Danilo Perez-Rivera, Heng Ma, Carla M. Mann, Michael Irvin, J. Gregory Pauloski, Logan Ward, Valerie Hayot-Sasson, Murali Emani, Sam Foreman, Zhen Xie, Diangen Lin, Maulik Shukla, Weili Nie, Josh Romero, Christian Dallago, Arash Vahdat, Chaowei Xiao, Thomas Gibbs, Ian Foster, James J. Davis, Michael E. Papka, Thomas Brettin, Rick Stevens, Anima Anandkumar, Venkatram

- Vishwanath, Arvind Ramanathan. GenSLMs: Genome-scale language models reveal SARS-CoV-2 evolutionary dynamics. International Journal of High Performance Computing Applications, 2023. ACM Gordon Bell Special Prize for COVID-19 Research
- 13 Justin Chiu\*, Yuntian Deng\*, Alexander Rush. Low-Rank Constraints for Fast Inference in Structured Models. NeurIPS 2021
  - 12 Yuntian Deng, Alexander Rush. Sequence-to-Lattice Models for Fast Translation. EMNLP 2021 Findings
  - 11 Keyon Vafa, Yuntian Deng, David Blei, Alexander Rush. Rationales for Sequential Predictions. EMNLP 2021 Oral
  - 10 Anton Bakhtin, Sam Gross, Myle Ott, Yuntian Deng, Marc’Aurelio Ranzato, Arthur Szlam. Real or Fake? Learning to Discriminate Machine from Human Generated Text. arXiv 1906.03351
  - 9 Thierry Tambe, En-Yu Yang, Zishen Wan, Yuntian Deng, Vijay Janapa Reddi, Alexander Rush, David Brooks, Gu-Yeon Wei. Algorithm-Hardware Co-Design of Adaptive Floating-Point Encodings for Resilient Deep Learning Inference. Design Automation Conference (DAC) 2020 Best Paper Award
  - 8 Yuntian Deng, David Rosenberg, Gideon Mann. Challenges in End-to-End Neural Scientific Table Recognition. ICDAR 2019
  - 7 Pengtao Xie, Yuntian Deng, Yi Zhou, Abhimanu Kumar, Yaoliang Yu, James Zou, Eric P Xing. Learning Latent Space Models with Angular Constraints. ICML 2017
  - 6 Guillaume Klein, Yoon Kim, Yuntian Deng, Jean Senellart, Alexander M Rush. OpenNMT: Open-Source Toolkit for Neural Machine Translation. ACL 2017 Best Demo Paper Award Runner-Up
  - 5 Zichao Yang, Zhiting Hu, Yuntian Deng, Chris Dyer, Alex Smola. Neural Machine Translation with Recurrent Attention Modeling. EACL 2017
  - 4 Xuezhe Ma, Yingkai Gao, Zhiting Hu, Yaoliang Yu, Yuntian Deng, Eduard Hovy. Dropout with Expectation-linear Regularization. ICLR 2017
  - 3 Hao Zhang, Zhiting Hu, Yuntian Deng, Mrinmaya Sachan, Zhicheng Yan, Eric P. Xing. Learning Concept Taxonomies from Multi-modal Data. ACL 2016
  - 2 Pengtao Xie, Yuntian Deng, Eric P. Xing. Diversifying Restricted Boltzmann Machine for Document Modeling. KDD 2015
  - 1 Zhiting Hu, Poyao Huang, Yuntian Deng, Yingkai Gao, Eric P. Xing. Entity Hierarchy Embedding. ACL 2015

#### Research Funding

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|                                                                                                                                                                               |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Laude Institute Slingshots: NeuralOS (USD 45,000 in-kind support)                                                                                                             | Jun 2026 |
| Google Research Award: “LLM-Guided Interactive Training of Biomedical Foundation Models on TPUs” (co-PI; USD 25,000 unrestricted gift and USD 75,000 in Google Cloud credits) | Apr 2026 |

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| CIFAR Canadian AI Safety Institute – AI Safety Solution Network: “Safeguarding Courts from Synthetic AI Content” (co-PI; CAD 700,000)                                                                                                                  | 2025–2027         |
| Canada Foundation for Innovation John R. Evans Leaders Fund: “Progressive Internalization of Skills for Advanced Language Model Reasoning” (PI; CAD 244,846 total project value, including CAD 96,000 CFI contribution)                                | 2024–2028         |
| Digital Research Alliance of Canada, Resources for Research Groups: “Neural OS: Simulating Operating Systems using Diffusion Models” (PI; RRG 2025 No. 5275, estimated in-kind value CAD 61,165; RRG 2026 ID 3797, estimated in-kind value CAD 28,772) | Jul 2025–Apr 2027 |
| Mitacs Business Strategy Internship: “Decentralized Electric Microgrid Optimization by Reinforcement Learning” (PI; CAD 10,000)                                                                                                                        | 2025              |
| Manulife Research Grant: “End-to-End Optimization of LLM-Based Multi-Agent Systems via Text Differentiation” (PI; CAD 30,000)                                                                                                                          | 2024–2026         |
| NSERC Discovery Grant: “Multi-Agent Communication and Collaboration for Specialized Large Language Models” (PI; RGPIN-2024-05178; CAD 41,000/year + CAD 12,500 Discovery Launch Supplement)                                                            | Aug 2024–Apr 2029 |
| University of Waterloo Starter Grant (CAD 150,000)                                                                                                                                                                                                     | Aug 2024–Aug 2029 |

## Media Coverage

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### Interviews and Direct Features

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|-------------------------------------------------------------------------------------------------------|--------------|
| The Canadian Press: Real or fake? Researchers to develop tool that would help courts spot AI evidence | Dec 5, 2025  |
| NZZ: What people are really asking ChatGPT? (English translation)                                     | Nov 30, 2024 |
| TechCrunch: Why is ChatGPT so bad at math?                                                            | Oct 2, 2024  |
| The Washington Post: What do people really ask chatbots?                                              | Aug 4, 2024  |

### Coverage of Coauthored Research

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|----------------------------------------------------------------------------------------------------------------|--------------|
| Tech Xplore: AI models stumble on basic multiplication without special training methods, study finds           | Dec 29, 2025 |
| Harvard Business School AI Institute: When Giants Stumble: What Multiplication Reveals about AI’s Capabilities | Oct 8, 2025  |
| Computerworld: Leaderboard illusion: How big tech skewed AI rankings on Chatbot Arena                          | May 2, 2025  |

## Workshop Organization

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Organizer of Latent & Implicit Thinking – Going Beyond CoT Reasoning Workshop Apr 27, 2026, ICLR

## Professional Service

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Editorial and Leadership: Co-Chief Technical Officer, ACL Rolling Review (ARR), Dec 2025–Present (technical infrastructure supporting 17,087 submissions in the May 2026 cycle); Action Editor, TACL, 2025–Present; Co-Chair, AAAI Undergraduate Consortium, 2027; Mentor Chair, ICLR 2026.

Area Chair / Senior Program Committee: Area Chair, ICLR (2024–2026), ICML (2026), NeurIPS (2026), ACL Rolling Review (2024–2026), COLM (2025–2026), EMNLP Demo (2026), LREC (2026), EMNLP GEM Workshop (2023), and IJCNLP–AAACL (Language Modeling, 2025); Senior Program Committee Member, AAAI 2027.

Reviewing: Invited Reviewer, Nature (2026); Reviewer, Schmidt Sciences Science of Trustworthy AI (2026); TACL, TMLR, Computational Linguistics, ACM AI Letters, ACM Computing Surveys, Foundations and Trends in Signal Processing, IEEE TNNLS, IEEE TIFS, ACM Transactions on Interactive Intelligent Systems, ACM Transactions on Information Systems, and Journal of Computer Science and Technology; Reviewer for ICML, NeurIPS, ICLR, ACL, ACL Rolling Review, EMNLP, AAAI, COLM, COLING, AAACL, EACL, NAACL, ACL Demo, NAACL Demo, EMNLP Demo, and various workshops and reproducibility challenges; Proposal Reviewer, Mitacs; Selection Committee Member, Vector Scholarship in Artificial Intelligence (2026–2027).

Conference Service: Oral Session Moderator, ICLR (2024, 2026); Oral Session Chair, EMNLP 2024; Co-host, ICLR 2026 Mentorship Session, Apr 23, 2026; Oral Session Volunteer, ACL 2021; Volunteer, EMNLP 2021 and ICML (2016, 2020); Volunteer Moderator, ACL 2020.

Recognition: Top 10% Reviewer, NeurIPS 2020; Best Reviewer (Top 10%), ICML 2021.

## Teaching Experience

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### Waterloo

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|-----------------------------------------------------------------|--------------|
| Instructor, CS 486/686: Introduction to Artificial Intelligence | Spring 2026  |
| Instructor, CS 486/686: Introduction to Artificial Intelligence | Spring 2025  |
| Guest Lecture, Computational Linguistics (Freda Shi)            | Feb 11, 2025 |
| Instructor, CS 886: Topics in Language Models                   | Fall 2024    |

### Harvard University

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|----------------------------------------------------------------------|-------------------|
| Head TA, Intro to Computational Linguistics and NLP (Stuart Shieber) | Aug 2021–Dec 2021 |
| Head TA, Intro to Computational Linguistics and NLP (Stuart Shieber) | Aug 2020–Dec 2020 |
| TA, Natural Language Processing (Alexander Rush)                     | Jan 2019–May 2019 |
| Lecture, Natural Language Processing – Translation (Alexander Rush)  | Feb 13, 2019      |

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|------------------------------------------------------------------------------------------------------------|-------------------|
| TA, Advanced Machine Learning (Alexander Rush)                                                             | Aug 2017–Dec 2017 |
| Cornell University                                                                                         |                   |
| Lab Material Preparation, Break Through Tech AI (Alexander Rush)                                           | Aug 2021          |
| Carnegie Mellon University                                                                                 |                   |
| TA, Probabilistic Graphical Models (Matthew Gormley, Eric Xing)                                            | Jan 2016–May 2016 |
| TA, Graduate Machine Learning (Ziv Bar-Joseph, Eric Xing)                                                  | Aug 2015–Dec 2015 |
| Community Service                                                                                          |                   |
| Co-Treasurer, Board of Directors, Bright Starts Early Learning Centre                                      | May 2025–May 2027 |
| Childcare Chair, NAACL 2022 Diversity & Inclusion Committee                                                | Apr–Jul 2022      |
| Leader, Harvard English Language Table                                                                     | Sep 2021–Jan 2023 |
| Mentor, Harvard Women in CS NLP Reading Group                                                              | Sep 2020–Sep 2023 |
| Volunteer Instructor of AP CS, Revere High School                                                          | Sep 2020–May 2021 |
| Internships                                                                                                |                   |
| NVIDIA (Anima Anandkumar, Weili Nie, Arash Vahdat, Chaowei Xiao)                                           | May–Dec 2022      |
| Facebook AI Research (Marc’Aurelio Ranzato, Arthur Szlam)                                                  | May–Dec 2019      |
| Bloomberg CTO Office (David Rosenberg, Gideon Mann)                                                        | Jan–Aug 2017      |
| UCSD (Charles Elkan)                                                                                       | Jul–Sep 2013      |
| Talks                                                                                                      |                   |
| Apple MLR Internal Reading Group: NeuralOS                                                                 | Aug 2026, Remote  |
| Vector Endless Summer School: NeuralOS                                                                     | Jun 2026, Remote  |
| NSERC CREATE Responsible AI consortium: NeuralOS                                                           | Mar 2026, Remote  |
| Mila/McGill NLP Reading Group: Implicit Reasoning in Language Models                                       | Oct 2025, Remote  |
| U Cambridge Language Technology Lab: Implicit Chain-of-Thought: Internalizing Reasoning in Language Models | Oct 2025, Remote  |
| Google DeepMind: Implicit Reasoning in Language Models                                                     | May 2025, Remote  |
| Colin Raffel Group: Implicit Reasoning in Language Models                                                  | Mar 2025, Toronto |
| U Oklahoma: Implicit Chain of Thought Reasoning                                                            | Jan 2025, Remote  |



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| NYU: Implicit Chain of Thought Reasoning                                  | Nov 2024, New York, NY        |
| Google DeepMind: Implicit Chain of Thought Reasoning                      | Jul 2024, Remote              |
| Queen's University: Implicit Chain of Thought Reasoning                   | Feb 2024, Remote              |
| UBC: Implicit Chain of Thought Reasoning                                  | Jan 2024, Vancouver, Canada   |
| AI2 Mosaic: WildChat                                                      | Sep 2023, Seattle, Washington |
| AI2 Semantic Scholar: Structure Modeling in Language Models               | Aug 2023, Seattle, Washington |
| NYU AI School 2023: Natural Language Processing                           | Jun 2023, New York, NY        |
| AI2 Mosaic: Structure Modeling in Language Models                         | Jun 2023, Seattle, Washington |
| UMass Amherst: Structural Coherence in Text Generation                    | May 2023, Remote              |
| Georgia Tech: Structural Coherence in Text Generation                     | Apr 2023, Atlanta, Georgia    |
| Harvard Kempner Institute: Structural Coherence in Text Generation        | Mar 2023, Remote              |
| U Alberta: Structural Coherence in Text Generation                        | Mar 2023, Edmonton, Canada    |
| TTIC: Structural Coherence in Text Generation                             | Feb 2023, Chicago             |
| U Waterloo ECE: Structural Coherence in Text Generation                   | Feb 2023, Remote              |
| Cornell Seminar in NLU: Structural Coherence in Text Generation           | Feb 2023, New York            |
| U Waterloo CS: Structural Coherence in Text Generation                    | Jan 2023, Waterloo, Canada    |
| U Chicago Rising Star: Model Criticism for Long-Form Text Generation      | Nov 2022, Chicago             |
| Princeton NLP: Model Criticism for Long-Form Text Generation              | Nov 2022, Princeton           |
| EMNLP 2021 Findings: Sequence-to-Lattice Models for Fast Translation      | Nov 2021, Remote              |
| OpenAI: Residual Energy-based Models for Text Generation                  | Apr 2021, Remote              |
| NeurIPS 2020: Cascaded Text Generation with Markov Transformers           | Dec 2020, Remote              |
| Baidu: Cascaded Text Generation with Markov Transformers                  | Jun 2020, Remote              |
| ICLR 2020: Residual Energy-based Models for Text Generation               | Apr 2020, Remote              |
| LinkedIn: Residual Energy-based Models for Text Generation                | Mar 2020, Remote              |
| Wayfair: Neural Encoder-Decoder Models                                    | Nov 2019, Boston              |
| EMNLP 2019: Neural Linguistic Steganography                               | Nov 2019, Hong Kong           |
| FAIR NLP Meeting: Energy-Based Models for Text Generation                 | Aug 2019, New York City       |
| NeurIPS 2018: Latent Alignment and Variational Attention                  | Dec 2018, Montreal, Canada    |
| The French National Center for Scientific Research: Variational Attention | Jul 2018, France              |

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| Association for Machine Translation in the Americas: OpenNMT   | Mar 2018, Boston            |
| Open-Source Neural Machine Translation Workshop: Image-to-Text | Mar 2018, Paris, France     |
| Tencent Social Network Group TSAIC: OpenNMT                    | Dec 2017, Shenzhen, China   |
| ICML: Image-to-Markup Generation                               | Aug 2017, Sydney, Australia |

## Panels

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|                                                                                                                                                               |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| KAIST at ICML                                                                                                                                                 | Jul 6, 2026, Seoul  |
| Panelist alongside Jaehong Yoon and Young Jin Kim. Discussed Vision, Multimodal AI & Agents.                                                                  |                     |
| Vector Institute's Remarkable 2025                                                                                                                            | Mar 2025, Toronto   |
| Panelist alongside Colin Raffel, Frank Rudzicz, Gerald Penn, Hongyang Zhang, Tracy Jenkin, and David Duvenaud. Discussed questions in NLP.                    |                     |
| ACL Mentorship: Sharing Learnings & Identifying Research Directions                                                                                           | Dec 2023, Singapore |
| Panelist alongside Mohit Bansal, Zhijing Jin, and Rada Mihalcea. Discussed promising research directions.                                                     |                     |
| ACL Mentorship: Preparing Your Grad School Application                                                                                                        | Oct 2023, Remote    |
| Participated in a mentorship session with Luke Zettlemoyer, Zhijing Jin, Nathan Schneider, and Oana Ignat, focused on preparing graduate school applications. |                     |
| NAACL 2022: Balancing Professional and Family Commitments                                                                                                     | Jul 2022, Remote    |
| Co-hosted with Shirley Hayati, discussing the balance between professional and family commitments.                                                            |                     |

## Thesis Committees

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- MASc Thesis Committee for Guilherme Campos (Waterloo, advised by Wojciech Golab and Tahsin Reza) Aug 12, 2026
- MMath Thesis Committee for Benjamin Schneider (Waterloo, advised by Wenhui Chen and Florian Kerschbaum) Aug 11, 2026
- MMath Thesis Committee for Arthur Chen (Waterloo, advised by Victor Zhong) Aug 4, 2026
- MMath Thesis Committee for Ruotian Wu (Waterloo, advised by Pascal Poupart) Aug 4, 2026
- PhD Thesis Committee for William Loh (Waterloo, advised by Pascal Poupart) Jul 21, 2026
- PhD Thesis Committee for Xueguang Ma (Waterloo, advised by Jimmy Lin) Jul 15, 2026
- MMath Thesis Committee for Hosna Oyarhoseini (Waterloo, advised by Jimmy Lin and Amir-Hossein Karimi) Jul 14, 2026
- PhD Comprehensive-II Committee for Xiang Li (Waterloo, advised by Jian Zhao and Kate Larson) Jun 26, 2026
- MMath Thesis Committee for Justin Xu (Waterloo, advised by Pascal Poupart) May 2026

- MMath Thesis Committee for Saarang Agarwal (Waterloo, advised by Mei Nagappan and Pengyu Nie) Apr 2026
- PhD Comprehensive-II Committee for Weiming Ren (Waterloo, advised by Wenhui Chen and Jimmy Lin) Apr 2026
- PhD Comprehensive-II Committee for Dongfu Jiang (Waterloo, advised by Wenhui Chen) Mar 2026
- PhD Comprehensive-II Committee for Cong Wei (Waterloo, advised by Wenhui Chen) Mar 2026
- PhD Comprehensive-II Committee for Dongfu Jiang (Waterloo, advised by Wenhui Chen) Jan 2026
- MMath Thesis Defense Committee for Yuansheng Ni (Waterloo, advised by Wenhui Chen) Nov 2025
- PhD Comprehensive-II Committee for Xueguang Ma (Waterloo, advised by Jimmy Lin) Oct 2025
- PhD Seminar Committee for Ronak Pradeep (Waterloo, advised by Jimmy Lin) Oct 2025
- MMath Thesis Defense Committee for Allen Zhang (Waterloo, advised by Anita Layton) Aug 2025
- MMath Thesis Defense Committee for Carter Blair (Waterloo, advised by Kate Larson and Edith Law) Jun 2025
- MMath Thesis Defense Committee for Achint Soni (Waterloo, advised by Sirisha Rambhatla and Charles Clarke) May 2025
- MMath Thesis Defense Committee for Chi-Chung Cheung (Waterloo, advised by Anita Layton) Dec 2024
- PhD Thesis Proposal Committee for William Loh (Waterloo, advised by Pascal Poupart) Aug 2024

#### PhD Seminar Attendance

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- Faculty Attendee, PhD Seminar for Cong Wei (Waterloo, advised by Wenhui Chen) Jul 27, 2026
- Faculty Attendee, PhD Seminar for Cong Wei (Waterloo, advised by Wenhui Chen) Jul 24, 2026
- Faculty Attendee, PhD Seminar for Weiming Ren (Waterloo, advised by Wenhui Chen) Jul 22, 2026
- Faculty Attendee, PhD Seminar for Dongfu Jiang (Waterloo, advised by Wenhui Chen) Jul 15, 2026
- Faculty Attendee, PhD Seminar for Weiming Ren (Waterloo, advised by Wenhui Chen) Jul 15, 2026
- Faculty Attendee, PhD Seminar for William Loh (Waterloo, advised by Pascal Poupart) Jun 15, 2026
- Faculty Attendee, PhD Seminar for William Loh (Waterloo, advised by Pascal Poupart) Jun 5, 2026
- Faculty Attendee, PhD Seminar for William Loh (Waterloo, advised by Pascal Poupart) Mar 2026

#### Open Source Projects

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|                                                                            |                   |
|----------------------------------------------------------------------------|-------------------|
| ProgramAsWeights (230 GitHub stars; 3,856 community-built neural programs) | Python            |
| NeuralOS (202 GitHub stars; 50,000+ demo visits)                           | Flask & WebSocket |
| Interactive Training (20 GitHub stars)                                     | Python            |
| WildChat-4.8M-Full                                                         | Dataset           |
| OpenAIWatch (8 Hugging Face likes)                                         | Python            |
| WildVisualizer (28 GitHub stars)                                           | Flask & D3.js     |
| Internalize CoT Step by Step (208 GitHub stars)                            | PyTorch           |
| Implicit Chain of Thought (149 GitHub stars)                               | PyTorch           |
| WildChat o1 Chatbot (450 Hugging Face likes)                               | PyTorch           |
| WildChat o1-mini Chatbot (258 Hugging Face likes)                          | PyTorch           |
| WildChat GPT-4 Chatbot (492 Hugging Face likes)                            | PyTorch           |
| ChatGPT Chatbot (416 Hugging Face likes)                                   | PyTorch           |
| Cascaded Generation (130 GitHub stars)                                     | PyTorch           |
| Neural Linguistic Steganography (214 GitHub stars)                         | PyTorch           |
| Variational Attention (327 GitHub stars)                                   | PyTorch           |
| Image-to-Markup Generation (1.3k GitHub stars)                             | LuaTorch          |
| OpenNMT-py (7k GitHub stars)                                               | PyTorch           |
| OpenNMT (archived) (2.4k GitHub stars)                                     | LuaTorch          |
| Attention OCR (1.1k GitHub stars)                                          | TensorFlow        |